

Poisson Regression

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Introduction

Poisson regression models count data with a log link, making it the canonical GLM for events per unit exposure: hospital admissions per month, mutations per kilobase, falls per resident-year. The log-linear structure means coefficients are interpretable as log-rate ratios, and the model accommodates varying exposures naturally through an offset term. The defining assumption — equidispersion, $\text{Var}(Y) = \mathbb{E}[Y]$ — is also its main limitation: real count data are typically overdispersed, and either quasi-Poisson, negative binomial, or zero-inflated extensions are then required.

Prerequisites

A working understanding of the Poisson distribution, generalised linear models, the log link, and the role of an offset for modelling rates rather than counts.

Theory

The model is $Y \sim \text{Poisson}(\mu)$ with $\log \mu = X^\top \beta$. Exponentiated coefficients are rate ratios: $\exp(\beta_j)$ multiplies the expected count per unit increase in X_j , all else equal. For rate-per-exposure problems, an offset of $\log(\text{exposure})$ converts the model into a rate model:

$$\log \mathbb{E}[Y] = \log(\text{exposure}) + X^\top \beta.$$

The offset's coefficient is fixed at 1, so the remaining β represent log-rate ratios per unit covariate change.

Equidispersion is the central assumption; if the variance exceeds the mean, standard errors are underestimated and inference is anti-conservative. Pearson chi-squared divided by residual degrees of freedom is the standard dispersion check; values much greater than 1 motivate quasi-Poisson or negative binomial.

Assumptions

Non-negative integer outcomes, independent observations, equidispersion (variance equal to mean), and a linear relationship between covariates and the log-rate.

R Implementation

```
library(broom)

# Simulated: event counts per patient-year
set.seed(2026)
d <- data.frame(
  age = rnorm(300, 60, 12),
  smoker = rbinom(300, 1, 0.3),
  exposure_years = runif(300, 0.5, 5)
)
d$events <- rpois(nrow(d),
```

```

lambda = d$exposure_years * exp(-2 + 0.02 * d$age + 0.6 * d$smoker))

fit <- glm(events ~ age + smoker, data = d, family = poisson,
           offset = log(exposure_years))
tidy(fit, conf.int = TRUE, exponentiate = TRUE)
glance(fit)

# Check overdispersion
dispersion <- sum(resid(fit, type = "pearson")^2) / fit$df.residual
dispersion

```

Output & Results

`tidy(fit, exponentiate = TRUE)` returns rate ratios and confidence intervals; `glance()` provides AIC, deviance, and other model-level summaries. The dispersion ratio close to 1 confirms Poisson is appropriate; values $\gg 1$ suggest overdispersion that needs different family handling.

Interpretation

A reporting sentence: “After adjusting for age (rate ratio 1.02 per year, 95 % CI 1.01 to 1.03), smokers had an event rate 1.81 times that of non-smokers (95 % CI 1.40 to 2.34, $p < 0.001$); the dispersion ratio of 1.05 is consistent with the Poisson assumption.” Always report rate ratios on the multiplicative scale and check dispersion.

Practical Tips

- Check the dispersion statistic immediately after fitting; if $\gg 1$, switch to quasi-Poisson (`family = quasipoisson`) for inflated SEs or to negative binomial (`MASS::glm.nb`) for explicit overdispersion modelling.
- Include an offset whenever exposure varies across observations; without it, coefficients absorb the exposure variation and are biased.
- Coefficients are on the log-rate scale; exponentiate for rate ratios in publication-friendly units.
- For excessive zero counts beyond what Poisson predicts, fit a zero-inflated or hurdle model (`pscl::zeroinfl, glmmTMB`).
- For count data with hierarchical structure (repeated measures, clustered observations), use Poisson mixed models (`glmer(family = poisson)` or `glmmTMB(family = poisson)`).
- Pearson residuals should look approximately uniformly scattered; systematic patterns suggest model misspecification (missing covariates, non-linear effects, overdispersion).

R Packages Used

Base R `glm()` with `family = poisson`; `MASS::glm.nb()` for negative binomial; `pscl::zeroinfl()` for zero-inflated Poisson; `glmmTMB` for mixed-effects Poisson, NB, and zero-inflated count models with cleaner overdispersion handling; `broom::tidy()` and `glance()` for tidy summaries.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI